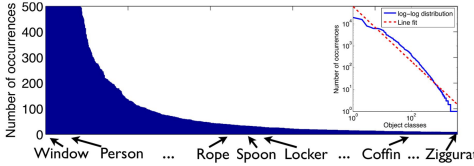
Optimization for Deep Learning (OPT4DL)

Abolfazl Hashemi, Purdue ECE

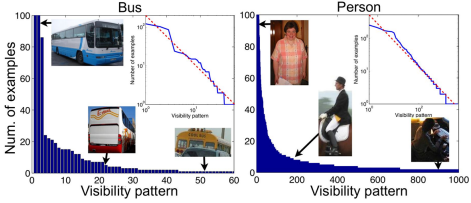
Memorization Phenomenon in Deep Learning

Motivation

- Differential Privacy: learn something useful about a population while not memorizing/learning something specific about an individual agent
- Noise-injection to Reduce sensitivity of our model to individual data points
- A natural trade-off between accuracy and privacy
- Is it always possible to learn something useful about a population while not memorizing/learning something specific about an individual agent?
- Role of data distribution: Long tailed vs Non-Long-Tail Distribution
- there are many small-frequency events (or values) that occur with relatively low probability



(a) The number of examples by object class in SUN dataset

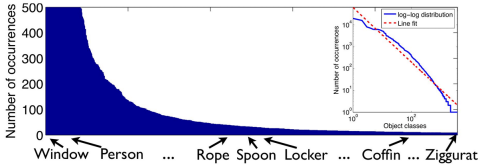


Long Tail Data Distributions

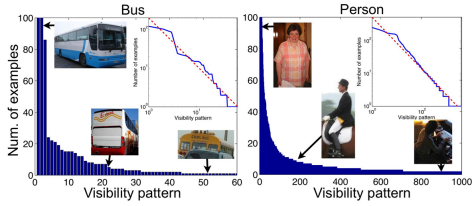
- A long-tail distribution
  is one where a large portion of the total probability mass is concentrated in the tail, which means there are many small-frequency events (or values) that occur with relatively low probability, but the tail of the distribution extends far out.
- Mathematically, the density follows Zipf's law

$$p(x) \propto x^{-s} \tag{1}$$

- In light-tail distribution such as Gaussian $p(x) \propto \exp(-x^2)$ the tail drops off much more quickly: Beyond a certain point, the probability of observing values far from the mean becomes extremely small.

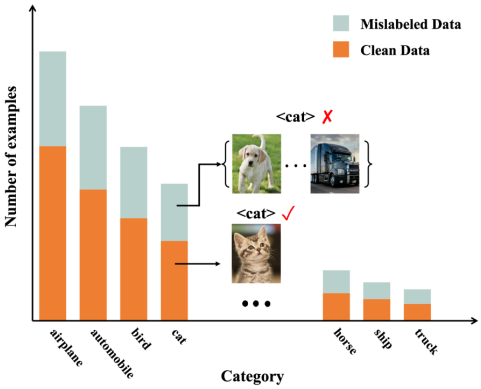


(a) The number of examples by object class in SUN dataset



Implication for Learning

- Long tail data points are atypical:
  they belong to subpopulations of size $\mathcal{O}(1/n)$
- If the
  model hopes to learn these subpopulations and generalize, it needs to memorize these samples
- Two questions arise:
  first what do we mean by memorization exactly?
  and secondly, are all atypical examples useful?
- Let us first
  answer the latter: No! atypical examples could include both "hard" examples as well as noisy (mislabeled, outliers, duplicated, etc.) examples



Memorization



- Memorization means model relies on specific training examples rather than learning general patterns applicable to unseen data

Definition

Let $\mathcal{A}$ a randomized algorithm (Such as SGD) applied to a training set S to learn a model h_S^w (e.g. a ResNet classifier) parameterized by w. Then, the memorization score for each data point is defined as

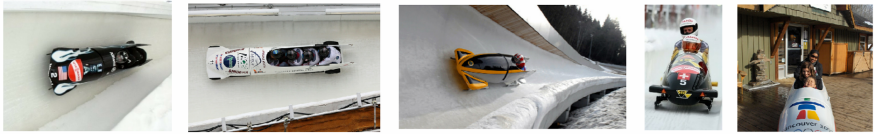
$$\mathrm{mem}(\mathcal{A}, S, z_i) = \mathbb{P}[h_S^w(x_i) = y_i] - \mathbb{P}[h_{S^{\setminus i}}^w(x_i) = y_i] \tag{2}$$

- measures the sensitivity of the model's predictions to the removal of individual data points from the training set.
- If removing a single example significantly alters the model's prediction for that example, the model is said to have memorized that example. Conversely, if the model's performance remains largely unchanged, it is considered more stable and less reliant on memorization.

Samples with Varied Memorization Scores

- Memorization Score for various data points from Class bobsled in ImageNet data set
- Low memorization score corresponds to typical and easy samples
- High memorization score corresponds to atypical (hard and also noisy) samples

0.0



0.5



1.0





Theorem

Let $e(\mathcal{A}, h^w, p_l)$ denote the expected generalization error of our model h^w trained by $\mathcal{A}$ for a long tailed distribution p_l. Let $e_{best}(h^w, p_l)$ denote the best achievable generalization error by any algorithm. Let $e_S(\mathcal{A}, 1)$ denote the number of examples that appear once (i.e. $freq_i = 1$) in the dataset S and are mislabeled by our model h^w trained by $\mathcal{A}$. Then,

$$e(\mathcal{A}, h^w, p_l) \geq e_{best}(h^w, p_l) + C \times e_S(\mathcal{A}, 1) \tag{3}$$

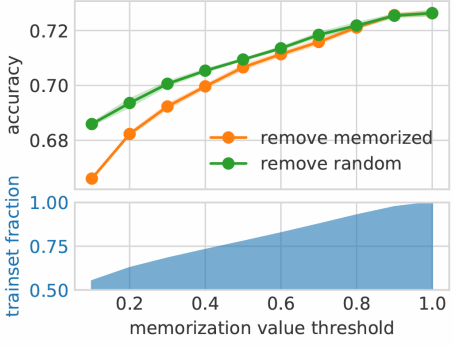
And crucially

$$e_S(\mathcal{A}, 1) = \sum_{i \in S;\ freq_i = 1} \mathbb{P}[h^w_{S \setminus i}(x_i) \neq y_i] - \mathrm{mem}(\mathcal{A}, S, z_i) \tag{4}$$

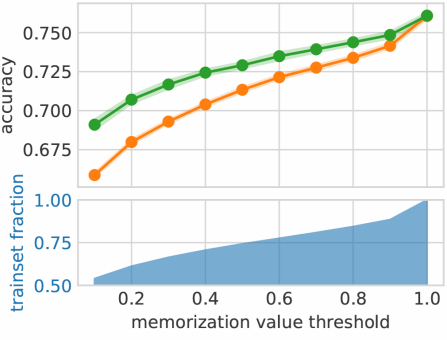
theorem implies to achieve the best performance, the model has to memorize atypical examples as $e_S(\mathcal{A}, 1)$ goes done as memorization goes up.

Removal Memorized Samples

- When removing samples with memorization score beyond a set threshold (e.g., 0.5) there is a significant drop in model's performance compared to random removal.
- Figure also demonstrates the long-tailed nature of data: more than 50% of samples have a memorization score below 0.1.



(a) ImageNet



(b) CIFAR-100

Memorization and Differential Privacy



- Recall, memorization score quantifies the sensitivity of the model's predictions to the removal of individual data points from the training set.
- Given the central role of sensitivity in the definition of both memorization and differential privacy, a natural question is about their relations

Theorem

Let $\mathcal{A}$ be a (ε, δ)-differentially private algorithm. Then with probability at least $1 - \delta$

$$\mathrm{mem}(\mathcal{A}, S, z_i) \leq 1 - e^{\epsilon}, \qquad \forall i \in S \tag{5}$$

- The theorem states that DP implies low memorization tendency (other direction?)
- As the privacy parameters go down, with increasing probability, $e^{\epsilon} \to 1$ and the memorization score goes down as well.
- This again highlights the privacy-accuracy tradeoff: In long tailed data distribution where memorization is essential, strong DP guarantees could hurt performance.

Input Loss Curvature and Memorization



- Recall, memorization score quantifies the sensitivity of the model's predictions to the removal of individual data points from the training set.
- The elegant definition of memorization is hard to compute and as mentioned it does not distinguish between truly hard examples and the mislabeled ones (no need to generalize well on a noisy sample).
- Therefore, there is a need for computationally efficient proxies that can also make this important distinction
- Our main toll to train a model is through minimizing the training loss. Thus, one could think of characterizing sensitivity by looking at loss $\ell(h_S^w, z_i)$ and its variation with respect to z_i
- Formally:

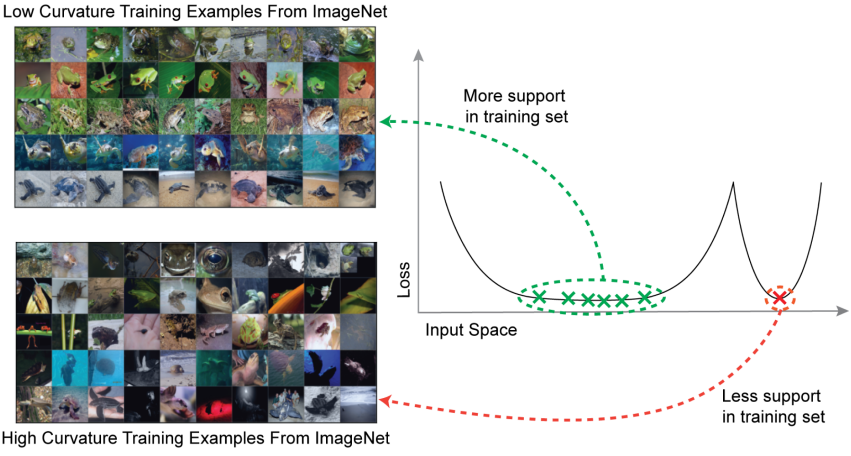
$$\mathrm{Curv}_w(z_i, S) = \mathrm{tr}(H_i) = \mathrm{tr}(\nabla^2_{z_i} \ell(h_S^w, z_i)) \tag{6}$$

Theorem

Under certain assumptions we can show $\mathrm{mem}(\mathcal{A}, S, z_i) \leq A\mathrm{Curv}_w(z_i, S) + B$

Title (Cont'd)

- curvature captures the local structure of the loss landscape:



Input Loss Curvature and Memorization (Cont'd)



- In practice, accumulated curvature $\sum_{t=1}^{T} \mathrm{Curv}_{w_t}(z_i, S)$ over the training process leads to higher correlation with memorization
- Curvature can be approximated efficiently using using Hutchinson's trace estimator and zeroth-order gradient estimation

$$\mathrm{Curv}_w(z_i, S)^2 \propto \frac{1}{M} \sum_{j=1}^{M} \|\nabla_{z_i} \ell(h_S^w, z_i + e_j) - \nabla_{z_i} \ell(h_S^w, z_i)\|^2, \qquad e_j \sim p_e \tag{7}$$

- Viewing curvature as second-order sensitivity, we can use first-order (input gradient $\|\nabla_{z_i} \ell(h_S^w, z_i)\|$) and zeroth-order (input loss) and their cumulative versions for more efficient memorization proxies, leading to broad family of differential input sensitivity (DIS) proxies
- One potential benefit of DIS is the ease to study memorization in unsupervised and generative models (active area of research)

Distinguishing Noisy and Hard Examples

- Both typical and atypical samples exhibit a low final loss. Noisy samples maintain a relatively high final loss.
- The average loss for typical samples remains low, whereas it is high for both atypical and noisy samples
- Could lead to the following criterion which is large for truly hard samples

$$\text{GoodMem} = \text{CumulativeDIS} - \text{FinalDIS} \tag{8}$$

